

Q-Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Protocol	Issue	Symptoms	Possible Causes	Recommended Solution
DNS	Incorrect DNS Server Address	Websites cannot be accessed by name, but can be accessed by IP address.	Wrong DNS server configured manually or through DHCP.	Configure the correct DNS server (e.g., 8.8.8.8 or the organization's DNS server) and renew the IP configuration.
DNS	DNS Server Unreachable	nslookup returns " DNS request timed out " or " Server failed ".	DNS server is offline, network connectivity issue, or firewall blocking DNS traffic.	Verify network connectivity, ensure the DNS server is running, and allow DNS traffic (port 53).
DNS	Incorrect DNS Record	A website opens the wrong page or cannot be found.	Incorrect or outdated DNS records, cached entries, or wrong hosts file configuration.	Update the DNS record, clear the DNS cache (ipconfig /flushdns), or correct the hosts file.
DHCP	Client Receives APIPA Address (169.254.x.x)	Client receives an IP address in the 169.254.x.x range and cannot access the network.	DHCP server unavailable, cable disconnected, router interface down, or DHCP service disabled.	Check physical connections, verify the DHCP server is running, enable the router interface, and renew the DHCP lease.
DHCP	Incorrect Default Gateway	Client receives an IP address but cannot communicate with devices outside the local network.	Wrong default-router value configured in the DHCP pool.	Correct the default gateway in the DHCP configuration and renew the client's IP address.
DHCP	DHCP Scope Exhaustion	New devices fail to obtain an IP address while existing devices continue to work.	All available IP addresses in the DHCP pool have been assigned.	Increase the DHCP address pool, shorten the lease time, or remove unused DHCP bindings.